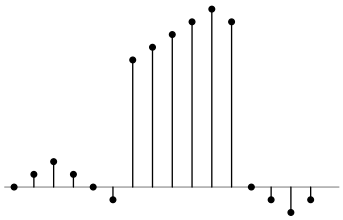
PAPER V / A WORKED VISUAL COMPANION

Three objects, three different questions

Main text: Sections 1, 6, 11 and 13. Read alongside the unchanged v1.1.0 manuscript.

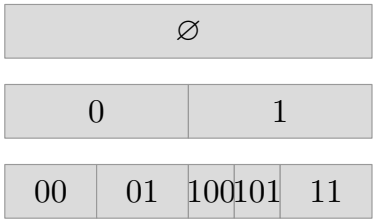
A packet calculus can reorganize a signal exactly, choose among finitely many representations, or describe a nonlinear coordinate geometry. These are not the same task. This appendix keeps one concrete object in view for each, so that the reader can compute rather than only follow arrows.

SIGNAL COORDINATES



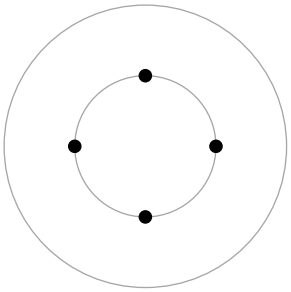
$f = s + j \quad \text{in } \mathbb{Q}^{16}$

FINITE TREE CHOICES



26 complete prunings

REFLECTION ORBIT



$\Gamma = \frac{1}{2} i^k$

A signal that can be checked by hand

Our main finite signal is $f = s + j \in \mathbb{Q}^{16}$. The background repeats

$$\tfrac{1}{4}(0, 1, 2, 1, 0, -1, -2, -1),$$

and $j[n] = 3$ at $6 \le n < 12$, with $j[n] = 0$ elsewhere. This is a discrete background model, not an assertion of continuum smoothness. Its energy is exactly 57. We use it for residue coordinates, Haar transforms, coefficient deletion, and known-packet cancellation.

A tree whose alternatives can all be seen

The full binary tree of depth three has only 26 complete prunings. We enumerate all of them, compute a unique mixed-depth optimum, and certify its stability with rational costs. The displayed objective is deliberately explicit, not claimed to be a universal signal-quality score.

A coordinate orbit with no numerical poles hidden

The four reflection points $\Gamma = i^k/2$ map to four exact impedances. A second orbit passes through infinity; homogeneous coordinates make that point an ordinary part of the construction.

> **Read this as:** Exact coordinates preserve information. An optimizer chooses coordinates. An estimator must still identify information not already supplied.

READING CONVENTIONS / KEEP THE TYPES VISIBLE

An atlas of the objects

Main text: notation crosswalk, p. 35; formal statement scopes, pp. 36–39.

Several symbols in the paper describe different realizations of the same operator language. The following distinctions are used throughout this appendix.

Object	Literal meaning	What it is not
S, δ_0	Sample shift and unit impulse	A finite truncation of every ℓ^2 signal
$P_{N,r} f$	A masked signal on the original lattice	The shorter decimated channel
$f_r[q]$	The value $f[Nq + r]$	An average of neighboring samples
D	One channel-index delay	Necessarily one original-sample delay
$E(D), R(D)$	Finite Laurent analysis and synthesis matrices	Values at one test frequency
W_α	The addressed Haar analysis operator	An edge detector by itself
$W_\alpha^* W_\alpha$	A projector on the original signal space	A scalar packet energy
Λ	A complete covering antichain of a finite tree	An arbitrary collection of nodes
$j, \widehat{j}$	Known and estimated discontinuity packets	Interchangeable hypotheses
Γ, z	Reflection coefficient and normalized impedance	Coefficients that can be linearly interchanged

Three arithmetic conventions

FIR examples use bilateral sequences with finite support and *zero extension*. Haar examples use vectors of length 16, with lengths divisible by the required powers of two. The three-tap smoother and nonlinear neighbor example use *periodic* indexing. These boundary conventions are stated where they enter; none is silently substituted for another. The polynomial variable t on $\mathrm{V}-20$ indexes coefficients. The delay symbol D acts on channels. The Smith variable z is impedance. Keeping them separate prevents a visual resemblance from becoming an invalid identity.

> **Read this as:** A correct formula needs the correct object, product, norm, and boundary convention. A familiar-looking matrix does not supply them automatically.

EXACT ADDRESSES / THE RUNNING SIGNAL

Decimation is a permutation, not a loss

Main text: Theorem 2.1 and Corollary 2.2, p. 6.

For four phases, write every sample address uniquely as $n = 4q + r$ with $0 \le r < 4$. Analysis stores $f[n]$ at channel address (r, q). Synthesis puts that exact value back at $4q + r$.

FOUR PHASES: EVERY ADDRESS APPEARS ONCE

$r = 0$	$n = 0:\ 0$	$n = 4:\ 0$	$n = 8:\ 3$	$n = 12:\ 0$	$n = 4q + r$
$r = 1$	$n = 1:\ \frac{1}{4}$	$n = 5:\ -\frac{1}{4}$	$n = 9:\ \frac{13}{4}$	$n = 13:\ -\frac{1}{4}$	
$r = 2$	$n = 2:\ \frac{1}{2}$	$n = 6:\ \frac{5}{2}$	$n = 10:\ \frac{7}{2}$	$n = 14:\ -\frac{1}{2}$	$-1 = 4(-1) + 3$
$r = 3$	$n = 3:\ \frac{1}{4}$	$n = 7:\ \frac{11}{4}$	$n = 11:\ \frac{13}{4}$	$n = 15:\ -\frac{1}{4}$	$-5 = 4(-2) + 3$
	$q = 0$	$q = 1$	$q = 2$	$q = 3$	

Every cell contains an original address and its rational sample value. Shading marks the known jump packet, not an estimated feature or a frequency band.

What has changed? Only where the values are stored

The complete tuple of channels contains sixteen values, just as the original vector does. A single decimated channel is shorter, but keeping all four loses no samples. The residue projector $P_{4,r} f$ is different: it keeps the original lattice and writes zeros outside phase r.

$$\mathcal{S}_4 \mathcal{A}_4 f = f, \qquad \|f\|_2^2 = \sum_{r=0}^{3} \|f_r\|_2^2 = 57.$$

The norm on the channel tuple is the sum-of-squares Hilbert norm, not the maximum of the channel norms. This distinction matters in the infinite-dimensional statement.

Negative addresses are ordinary addresses

Euclidean division, rather than truncation toward zero, gives $-1 = 4(-1) + 3$ and $-5 = 4(-2) + 3$. The finite tests exercise both positive and negative sample addresses. The algebraic identity applies before any plotting window or finite buffer is chosen.

> **Read this as:** Downsampling one channel discards information. Splitting into all residue channels is an invertible address change.

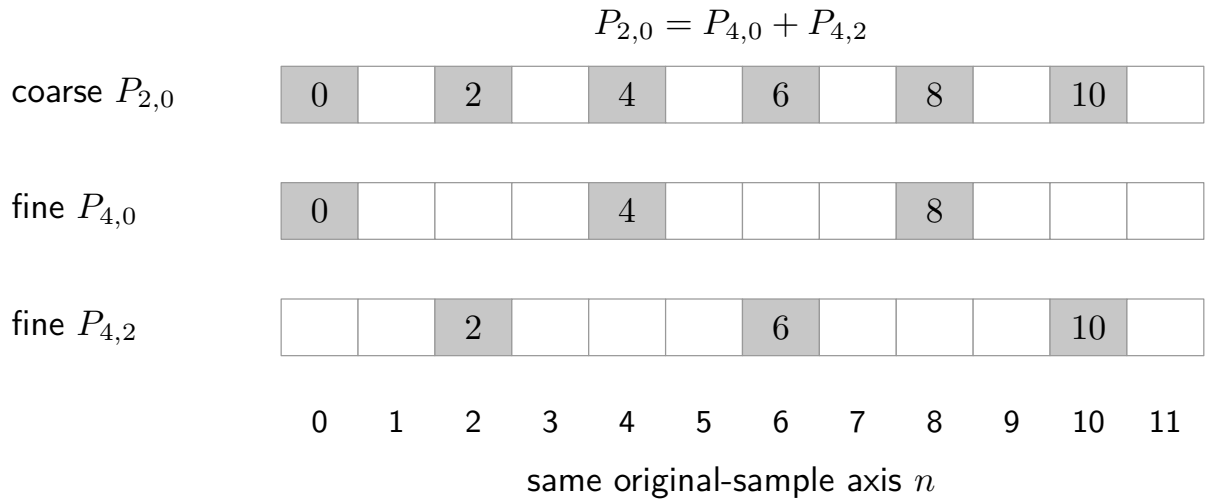
EXACT REFINEMENT / WHAT CAN BE ADDED

Refine packets on a common lattice

Main text: Theorem 2.3 and its coordinate convention, p. 8.

An even sample belongs to exactly one of residues 0 and 2 modulo four. Therefore the coarse even mask is the sum of two disjoint fine masks:

$$P_{2,0} = P_{4,0} + P_{4,2}.$$



The common horizontal axis is the original sample index. Empty cells denote masked-out addresses. There are no interpolated samples between the displayed cells.

Why the coordinate convention matters

The decimated channel $f_0[q] = f[2q]$ has its own index q. At modulus four the descendants are $f[4q]$ and $f[4q+2]$. These shorter arrays must be interleaved to recover the coarse channel. Adding them entrywise instead produces sums of different original samples. For example, at the first descendant address the values $f[0] = 0$ and $f[2] = 1/2$ are two separate values. Their sum $1/2$ cannot tell us both of them. Exact refinement is a disjoint partition, not an averaging rule or a new measurement.

The general formula

When N divides M,

$$P_{N,r} = \sum_{\substack{0 \le a < M \\ a \equiv r \pmod{N}}} P_{M,a}.$$

The divisibility hypothesis ensures that every fine residue class lies entirely inside one coarse class. Without it, this particular descendant relation is not the right refinement rule.

> **Read this as:** Finer coordinates can preserve the object exactly. Compressing their values to averages is an additional operation, with an additional information-loss question.

FIR TRANSFER / A FOUR-TAP EXAMPLE

A filter becomes a matrix of actual delays

Main text: Theorem 3.1, pp. 8–9.

Take the causal filter $h = (1, 2, -1, 1)$, with taps at $0, 1, 2, 3$, acting by $y[n] = \sum_q h[q] f[n-q]$. Resolve the input into even and odd channels. Let D denote one delay in the *channel* index.

$$h = (1, 2, -1, 1), \qquad D : \; u[n] \mapsto u[n-1]$$

	input 0	input 1
output 0	$1 - D$	$D^2 + 2D$
output 1	$D + 2$	$1 - D$

$$y_0 = (1 - D)x_0 + (2D + D^2)x_1$$

Reading one matrix entry

The odd-input contribution to the even output is $(2D + D^2)f_1$. The tap at original delay one reaches the preceding odd channel sample; the tap at delay three reaches the odd channel two positions back. These are different delays despite coming from the same input phase. The two channel equations are

$$y_0[n] = f_0[n] - f_0[n-1] + 2f_1[n-1] + f_1[n-2],$$
$$y_1[n] = 2f_0[n] + f_0[n-1] + f_1[n] - f_1[n-1].$$

Interleaving gives exactly the direct convolution, including its boundary samples under zero extension.

The general address calculation

For N phases, the block from input s to output k has kernel $h_{ks}[m] = h[Nm + k - s]$. This comes from the exact identity $Nn + k - q = N(n - m) + s$. It is a finite reindexing of taps, so no numerical transform of the input is needed. Evaluation of a bilateral transform is a separate step with the summability hypotheses stated in the manuscript.

> **Read this as:** The matrix does not approximate the filter. Its entries record which input phase is used, how far back to read, and what coefficient to multiply.

DELAY UNITS / AN IMPULSE MAKES THE DIFFERENCE VISIBLE

One sample is not one channel step

Main text: Theorem 4.1 and the original-sample delay formula, p. 10.

A one-sample delay swaps the two phases and wraps one of them into the preceding channel index. Its matrix is $J(D)$, not DI_2.

one original sample

0	D
1	0

$$J(D)^2 =$$

two original samples

D	0
0	D

$\delta_0 \mapsto \delta_1$

$\delta_0 \mapsto \delta_2$

$$y_0[n] = x_1[n-1], \quad y_1[n] = x_0[n]$$

channel delay 1 at $N = 2$ is original-sample delay 2

Trace one impulse

The input δ_0 has a single nonzero value in even channel address zero. Under $J(D)$ it appears in odd channel address zero: this is the original sample δ_1. Under DI_2 it stays even but moves to channel address one: this is δ_2. Thus

$$J(D)^2 = DI_2, \qquad H(D) = I_2 + 2J(D) - J(D)^2 + J(D)^3$$

recovers the previous page's four-tap transfer matrix. The same delay algebra explains both the off-diagonal phase swap and the channel powers.

Signed delays need the same care

For an original-sample delay d at modulus N,

$$y_k[n] = f_{(k-d) \bmod N}\left[n + \left\lfloor \frac{k-d}{N} \right\rfloor\right].$$

This includes advances when $d < 0$. The finite verification checks the formula on signed sample addresses. A bank with product $D^q I_N$ delays the original signal by Nq samples, not q samples.

> **Read this as:** Always state the unit of a delay. A phase permutation and a channel delay together can represent one original sample.

FILTER BANKS / WHY A FREQUENCY SPOT-CHECK IS INSUFFICIENT

Perfect reconstruction is an operator identity

Main text: Theorem 4.1 and its impulse proof, p. 10.

Consider the finite Laurent bank

$$E(D) = \begin{pmatrix} 1 & D \\ 0 & 1 \end{pmatrix}, \qquad R(D) = \begin{pmatrix} 1 & -D \\ 0 & 1 \end{pmatrix}.$$

Its synthesis cancels the exact delayed cross-channel contribution introduced by analysis.

$E(D)$

1	D
0	1

$R(D)$

1	$-D$
0	1

$$R(D)E(D) = I$$

every impulse is recovered

$$B(D) = \operatorname{diag}(D, 1): \quad B(1) = I, \quad B(D) \neq I$$

testing one frequency cannot detect a channel delay

Two inverses, all inputs

Direct polynomial multiplication gives $RE = ER = I_2$. Since D acts as a shift, these identities imply perfect reconstruction on every bilateral channel sequence. With finite kernels, composition requires only finitely many operations at each address. The converse uses impulses. Insert an impulse in one input channel. Each output channel lists every coefficient of the corresponding transfer polynomial. If all such responses vanish for $RE - I$, every coefficient is zero; hence the polynomial matrix itself is zero.

A deliberately misleading test

The matrix $B(D) = \operatorname{diag}(D, 1)$ satisfies $B(1) = I_2$. At that one transform value it appears perfect. In the time domain, however, it delays only the first channel. An impulse in that channel immediately distinguishes it from the identity. This is why a small set of successful frequency evaluations is not automatically a proof of the full Laurent identity. A separate degree bound and a sufficient interpolation argument would be needed to turn finitely many evaluations into such a certificate.

> **Read this as:** Exact reconstruction concerns every coefficient of the transfer operator, not just an attractive response curve or one evaluation point.

HAAR ENERGY / EDGE ALIGNMENT

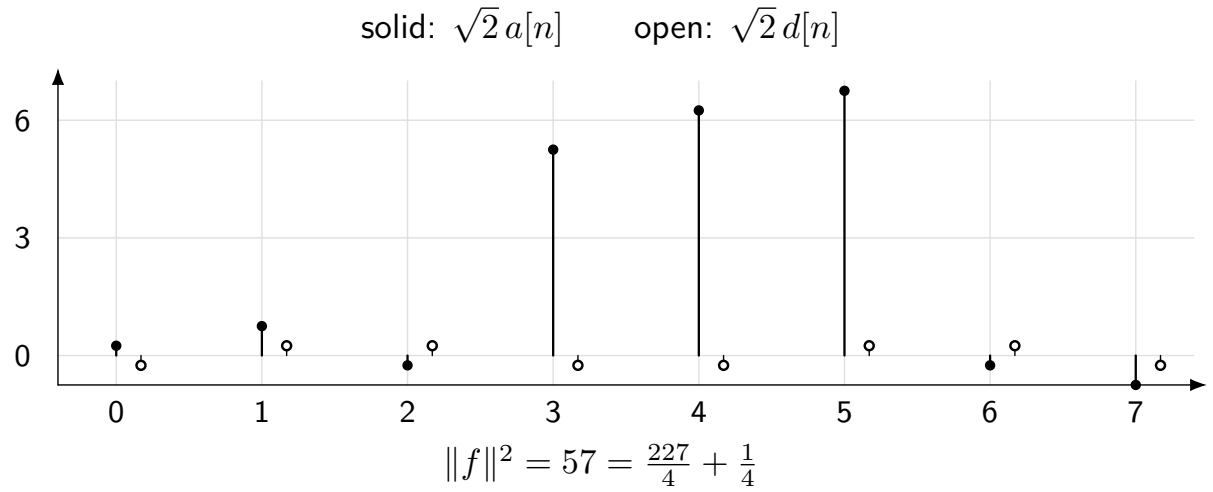
A jump can hide from the first Haar split

Main text: Corollary 4.2 and the literal channel formulas, pp. 10–11.

For the running signal, Haar analysis gives

$$a[n] = \frac{f[2n] + f[2n+1]}{\sqrt{2}}, \qquad d[n] = \frac{f[2n] - f[2n+1]}{\sqrt{2}}.$$

The plot scales both channels by $\sqrt{2}$ so that all displayed ordinates are rational.



The exact energy calculation

The low channel has energy 227/4, and the detail channel has energy 1/4. Their sum is 57, exactly the input energy. The normalization is essential: unnormalized sums and differences would multiply the total energy by two.

Why is the first detail channel so small?

The jump starts at sample 6 and ends just before sample 12. Both are boundaries *between* the pairs $(0,1),(2,3),\ldots,(14,15)$. Inside every pair the jump contributes either $(0,0)$ or $(3,3)$, so its first-level difference is zero. The small detail values come entirely from the discrete background. This does not mean the edge is absent from the transform. The next split of the low channel compares neighboring pairs, and those comparisons do cross the jump. The next pages expose that contribution explicitly.

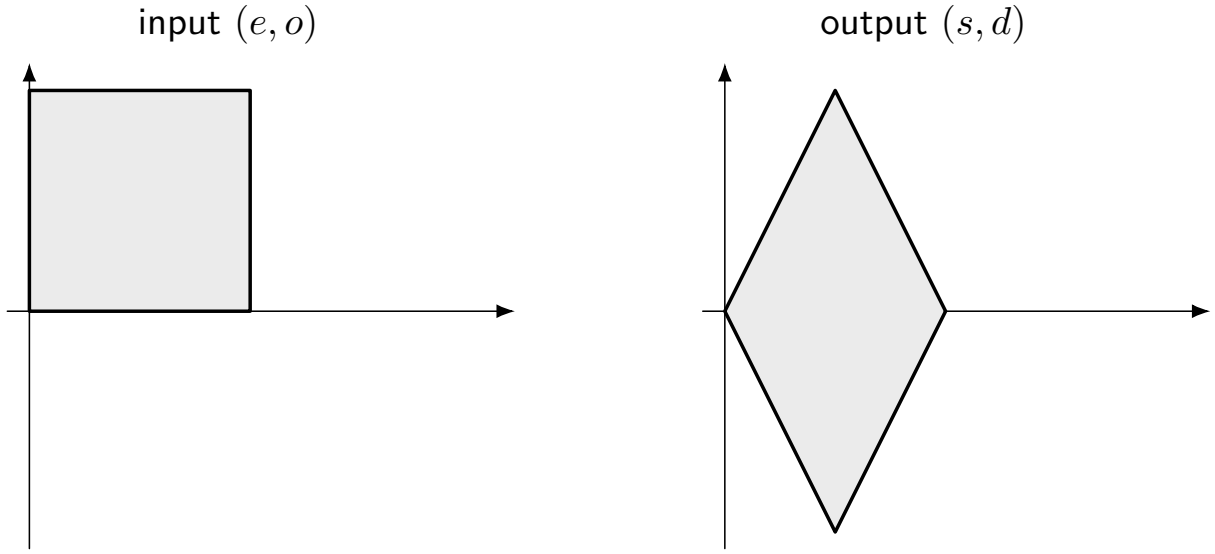
> **Read this as:** Haar energy is exact, but a particular detail level is not a universal edge detector. Edge visibility depends on alignment, scale, and the chosen packets.

PREDICT–UPDATE / INVERTIBILITY VERSUS ISOMETRY

Lifting preserves information, not energy

Main text: Theorem 4.3, pp. 11–12.

For an even/odd pair (e, o), form the prediction residual $d = o - e$, then update $s = e + d/2$. The inverse reads $e = s - d/2$, followed by $o = d + e$.



$$(2,6): \quad 2^2 + 6^2 = 40 \qquad (4,4): \quad 4^2 + 4^2 = 32$$

The square $[0,4]^2$ and its exact image under the lifting matrix have the same area. The separate pair $(2,6)$ illustrates that Euclidean energy need not be preserved.

$$\begin{pmatrix} s \\ d \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e \\ o \end{pmatrix}, \qquad \begin{pmatrix} e \\ o \end{pmatrix} = \begin{pmatrix} 1 & -1/2 \\ 1 & 1/2 \end{pmatrix} \begin{pmatrix} s \\ d \end{pmatrix}.$$

What determinant one does establish

The matrix is invertible, and in these real coordinates it preserves oriented area. The predict and update operations can be undone in reverse order. Applied to each pair, the construction gives exact finite-signal reconstruction.

What determinant one does not establish

The pair $(2,6)$ becomes $(4,4)$. Its squared Euclidean norm changes from 40 to 32. The lifting matrix is not unitary. Energy accounting must therefore use its actual coordinate metric or a separately normalized transform; invertibility alone cannot replace the Haar energy identity. All values in this example are rational. Integer-to-integer lifting with a rounding rule is a different algorithm. Its inverse and error conventions require their own specification and are not silently inferred from the displayed rational matrix.

> **Read this as:** Perfect reconstruction, determinant one, and norm preservation are different properties. This transform has the first two, not the third.

ACTUAL WAVELET PACKETS / FOUR RATIONAL CHANNELS

The second level resolves the hidden change

Main text: Theorem 5.1, p. 12.

Apply Haar analysis to *both* children. A binary address records the chronological low/detail choices, with 0 for low and 1 for detail. Each depth-two channel contains four coefficients.

	coefficient positions within each packet				energy
00	$\frac{1}{2}$	$\frac{5}{2}$	$\frac{13}{2}$	$-\frac{1}{2}$	49
01	$-\frac{1}{4}$	$-\frac{11}{4}$	$-\frac{1}{4}$	$\frac{1}{4}$	$\frac{31}{4}$
10	0	0	0	0	0
11	$-\frac{1}{4}$	$\frac{1}{4}$	$-\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

$$\sum_{\alpha\in\{0,1\}^2} W_\alpha^* W_\alpha = I_{16}$$

The edge that first-level detail missed

Packet 01 is the detail child of the first low channel. It contains $-11/4$, the large change between neighboring pair averages. Packet 10 vanishes for this particular signal because the first-level differences alternate in cancelling pairs. Neither feature is a universal identity for all inputs. The packet energies are

$$49, \qquad 31/4, \qquad 0, \qquad 1/4,$$

which again sum to 57. Two factors of $1/\sqrt{2}$ explain why the depth-two coefficients are rational halves and quarters.

Coefficients and projectors are different objects

$W_\alpha f$ is the short coefficient vector. The original-space component is $W_\alpha^* W_\alpha f$. Summing these synthesized components, not the short arrays entrywise, reconstructs f:

$$f = \sum_{|\alpha|=2} W_\alpha^* W_\alpha f.$$

The finite verification constructs the actual analysis matrices and checks their product with their adjoints. The paper's ℓ^2 statement additionally establishes boundedness and the existence of the literal Hilbert adjoints.

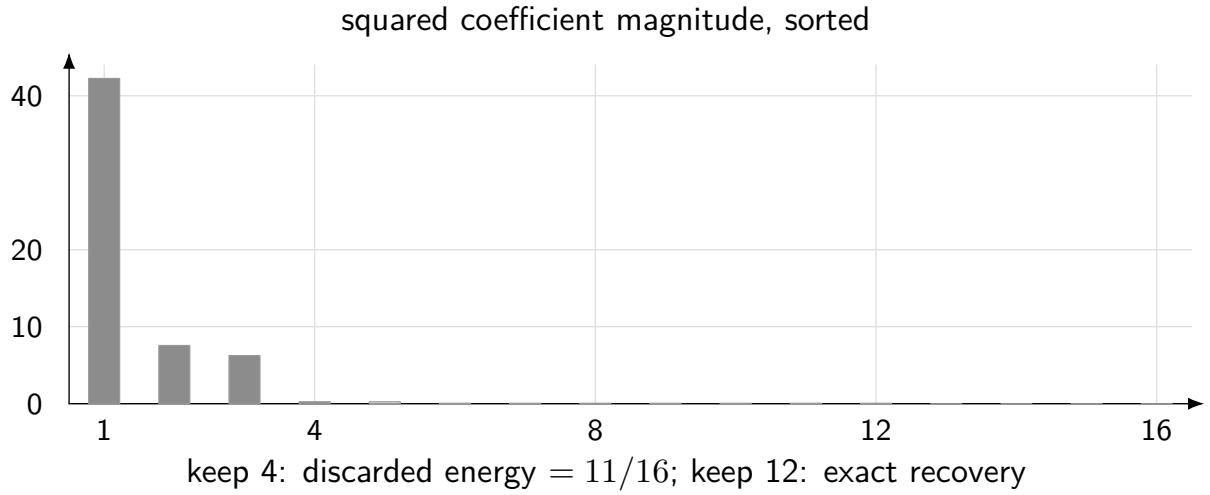
> **Read this as:** A packet address specifies an operator word. To add different packets as signals, synthesize them back into the common signal space first.

SPARSITY / AN EXACT ERROR LEDGER

Deleting coefficients is the approximation step

Main text: Section 10 and its basis-selection boundary, pp. 20–21.

The complete depth-two transform is exact. Approximation begins only when we discard some of its sixteen coefficients. Order them by squared magnitude, with a deterministic address-and-position rule for ties.



Every discarded square has a meaning

In this fixed orthonormal basis, the squared reconstruction error equals the sum of the squared discarded coefficients. For our running signal:

kept coefficients k	1	2	4	8	12	16
squared error	59/4	115/16	11/16	1/4	0	0

The four zero coefficients belong to packet 10, so twelve coefficients suffice for exact reconstruction here. This is a property of the selected signal and basis, not a guarantee that all edge signals need only twelve coefficients.

Do not compare incompatible budgets

A cutoff m in a one-dimensional signed Fourier spectrum means $2m + 1$ bins, including zero. A budget of k largest coefficients is a different rule. Real Fourier synthesis may also require conjugate-pair closure. Neither count alone equals a bit-storage budget once amplitudes, phases, indices, or a selected tree must be encoded. The manuscript's Fourier-versus-Haar experiment is a finite comparison, not a proof that one basis dominates every signal class. Here we isolate the part that can be checked exactly: reconstruction error within one fixed finite basis.

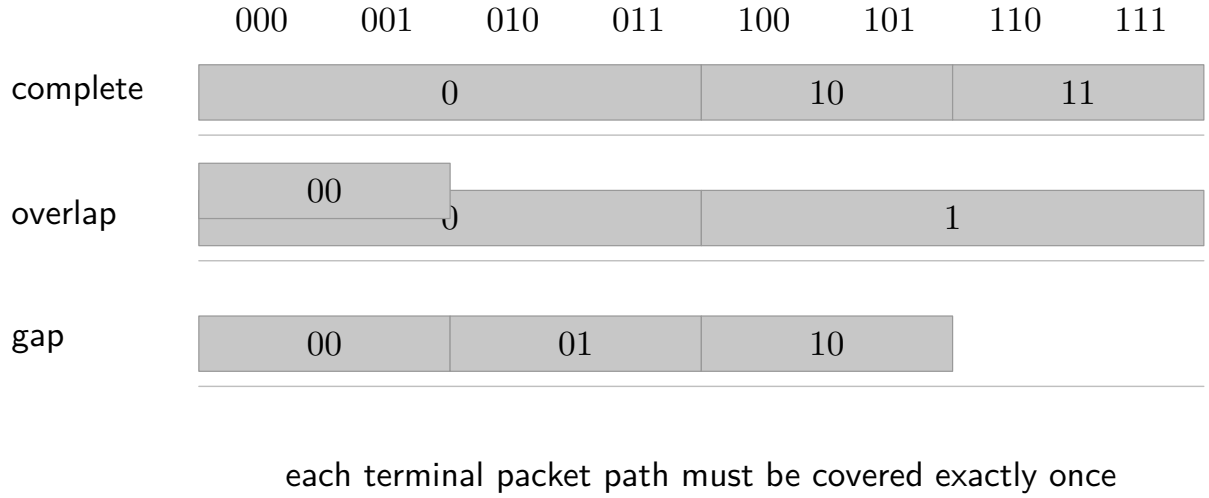
> **Read this as:** Changing coordinates creates no error. Throwing away coefficients does, and orthogonality makes that error exactly accountable.

COMPLETE PRUNINGS / NO GAPS AND NO DOUBLE COUNTING

A mixed-depth basis must cover every branch

Main text: Definition 11.1 and Theorem 11.2, p. 21.

A complete pruning chooses nodes of a finite refinement tree so that every root-to-terminal path meets exactly one selected node. The horizontal positions below are *terminal packet addresses*, not time intervals in the original signal.



Three candidate selections

The complete selection $\{0, 10, 11\}$ covers every terminal branch once. The selection $\{0, 00, 1\}$ covers the 00 branch twice: it is not an antichain. The selection $\{00, 01, 10\}$ misses the 11 branch entirely. For actual Haar packets, set $P_v = W_v^* W_v$. Parent-to-child conservation gives

$$P_v = P_{v0} + P_{v1}.$$

Replacing unselected parents by their children until the complete pruning is reached proves $\sum_{v \in \Lambda} P_v = I$. This is why the coefficients at different depths still form an exact coordinate system.

An exhaustive finite check

At depth three the pruning count obeys $a_0 = 1$ and $a_{d+1} = 1 + a_d^2$, giving $1, 2, 5, 26$. The new example checks all 26 covering conditions and all 26 actual 16×16 projector-sum identities. These finite checks illustrate, but do not replace, the universal tree theorem.

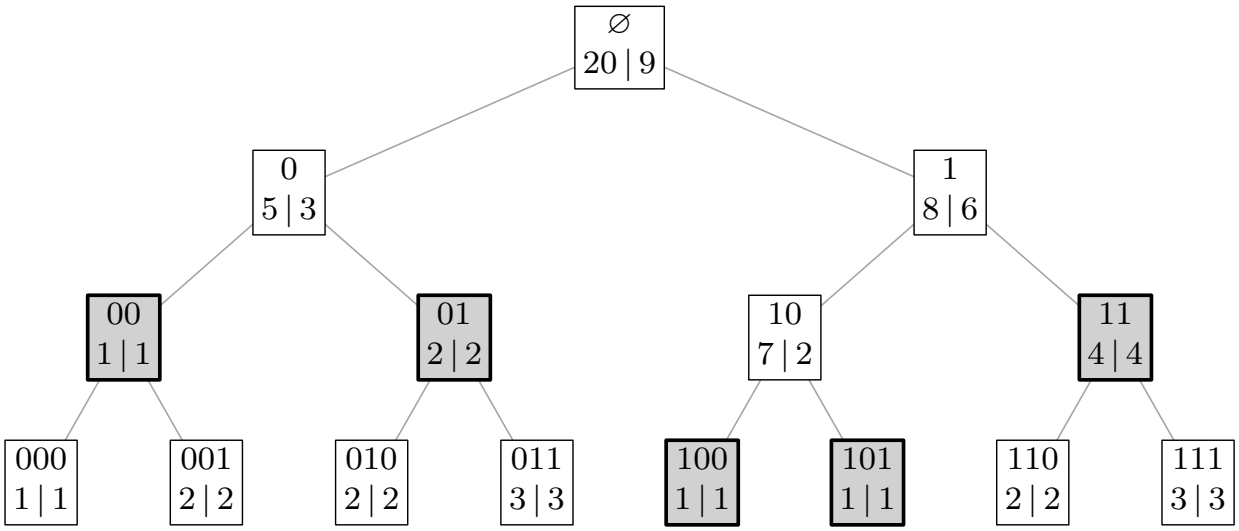
> **Read this as:** Mixed depth is allowed. Missing a terminal path or selecting an ancestor and its descendant is not.

ADDITIVE COSTS / EXACT DYNAMIC PROGRAMMING

Solve every alternative without searching every alternative

Main text: Theorems 11.3–11.4, pp. 21–23.

Give every node the rational stop cost shown below. A node may either stop and pay its own cost, or refine and pay the already optimized costs of its children. The shaded nodes form the selected complete pruning.



node labels: stop cost c_v | optimal subtree cost $V(v)$

$$V(v) = \min\{c_v,\ V(v0) + V(v1)\}, \qquad V(v) = c_v \text{ at a terminal node.}$$

Read the calculation from the bottom upward

At 10, the children cost $1+1 = 2$, better than stopping for 7. At 11, stopping costs 4, better than $2+3 = 5$. The right subtree therefore costs 6, better than its stop cost 8. The left subtree costs $1 + 2 = 3$, better than 5. At the root, $3 + 6 = 9$ beats 20. The unique optimum is

$$\Lambda_* = \{00, 01, 100, 101, 11\}, \qquad J(\Lambda_*) = 9.$$

All 26 complete prunings are separately enumerated and give the same minimum. At equality, the rule stops, making the selector deterministic. A singleton tree needs no comparison. The costs here are supplied numbers. The theorem optimizes that stated additive objective; it does not prove those numbers encode the best denoising criterion. Coupled costs that cannot be written as a sum of selected-node costs need a different optimizer.

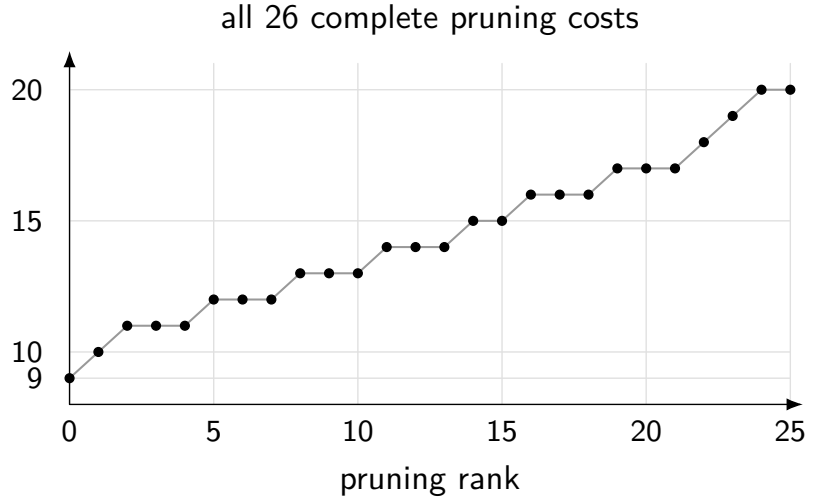
> **Read this as:** Global optimality follows from the exact stop-or-refine decomposition of a complete pruning, not from a greedy assumption.

A GLOBAL MARGIN / NOT A LOCAL HEURISTIC

How far can the costs move before the tree changes?

Main text: Theorem 11.3 and Corollary 11.5, pp. 22–23.

The best pruning on the previous page costs 9; the cheapest competitor costs 10. Its exact global gap is therefore $\Gamma = 1$. The plate lists every complete-pruning cost in sorted order, giving each pruning its own horizontal rank.



$$\Gamma = 10 - 9 = 1$$

$$E(\varnothing) = 8/32$$

$$2E = 1/2 < \Gamma$$

selector certified stable

Propagate an error budget as carefully as a cost

Suppose every node cost changes by at most $\varepsilon_v = 1/32$. A pruning may select as many as eight leaves, so its total cost can change by at most $8/32 = 1/4$. The recursive envelope proves this without guessing which pruning will be selected:

$$E(v) = \max\{\varepsilon_v,\ E(v0) + E(v1)\}, \qquad E(\varnothing) = \tfrac{1}{4}.$$

For any competitor Λ, the changed cost difference is bounded below by

$$J'(\Lambda) - J'(\Lambda_*) \geq \Gamma - 2E(\varnothing) = \tfrac{1}{2} > 0.$$

Thus the optimum is unchanged for *every* permitted simultaneous perturbation, not just a sampled noise realization. The envelope is over all feasible prunings, not just the five nodes in the current optimum.

Strict inequality has content

If two competing costs are 9 and 10, moving them to 9.5 and 9.5 removes uniqueness. Equality in $2E = \Gamma$ is therefore not a unique-selector certificate. In a probabilistic use, the node bounds must hold on one joint event; separate nodewise probabilities do not automatically give its probability.

> **Read this as:** The exact selector is stable because its global gap exceeds the total worst-case movement of both competing costs.

AN EXACT IDENTITY / A CONDITIONAL INPUT

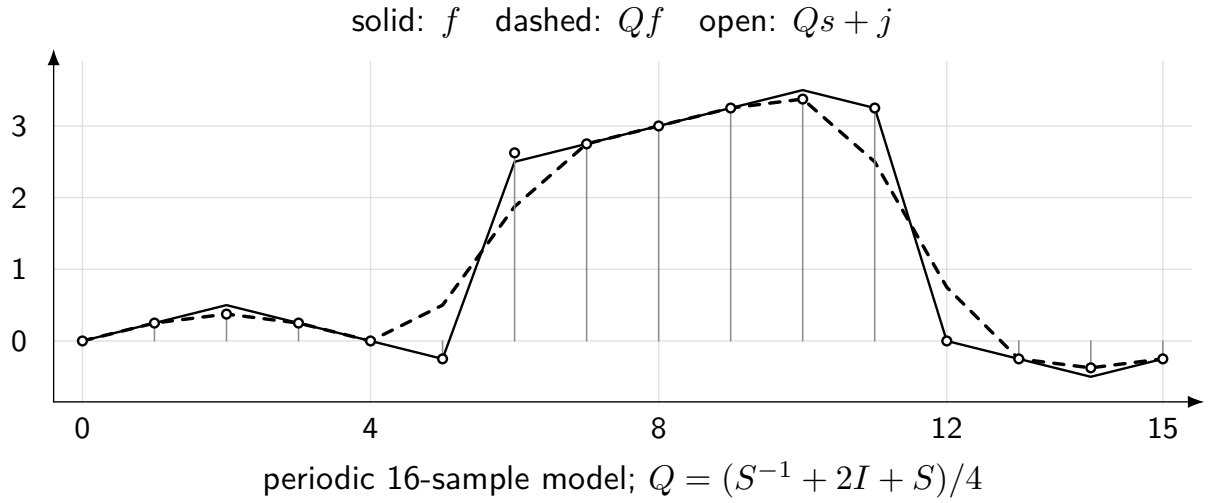
Retain the jump; approximate only the other packet

Main text: Theorem 6.1, Corollary 6.2, and Boundary 6.3, pp. 12–13.

Return to the running signal $f = s + j$. For a concrete finite calculation, use the periodic three-tap smoother

$$(Qx)[n] = \tfrac{1}{4}x[n-1] + \tfrac{1}{2}x[n] + \tfrac{1}{4}x[n+1], \qquad n \in \mathbb{Z}/16\mathbb{Z}.$$

The plots compare smoothing the full signal with retaining its supplied jump packet. Every plotted sample is a rational number computed from the displayed formula.



The cancellation is the theorem

Define $\widetilde{f} = Qs + j$. Then, sample by sample,

$$\boxed{f - \widetilde{f} = s - Qs.}$$

By contrast, linearity gives

$$f - Qf = (s - Qs) + (j - Qj).$$

The second error includes a new term caused by smoothing the discontinuity. In this finite example that term is concentrated near the two jump boundaries. The identity itself requires no linearity of the approximation applied to s; linearity is used only in the comparison with Qf.

The input is not discovered by subtraction

The data here include the correct j. Replacing it with a wrong $\widehat{j}$ retains the error $j - \widehat{j}$, even when s is approximated perfectly. Multiple amplitudes and overlapping shapes are allowed if their full sum is supplied, but the identity does not estimate that sum from observations. Our finite periodic smoother is an illustration, not a new continuum approximation theorem. Its chosen boundary convention is part of the experiment.

> **Read this as:** Known-edge reconstruction removes the jump from the approximation problem. It does not remove the need to identify the jump.

A CONTINUOUS PULSE / EXACT ENDPOINT VALUES

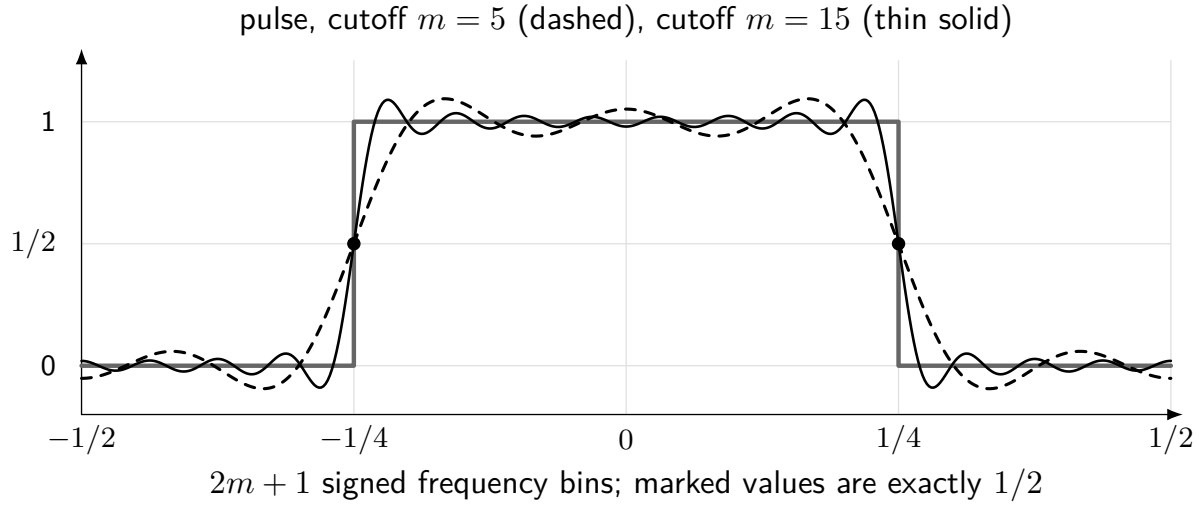
Why a Fourier cutoff cannot reproduce this jump pointwise

Main text: Sections 6 and 8, pp. 12–15.

Use a separate period-one example on $[-1/2, 1/2)$:

$$j(t) = \mathbf{1}_{\{|t|<1/4\}}, \qquad S_m j(t) = \tfrac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{m} \frac{\sin(\pi n/2)}{n} \cos(2\pi n t).$$

This is the exact finite Fourier formula. The drawn curves sample it numerically; the endpoint calculation below is symbolic.



One value is fixed at every cutoff

At $t = \pm 1/4$, every summand vanishes because $\sin(\pi n/2)\cos(\pi n/2) = 0$ for integer n. Hence

$$S_m j(\pm 1/4) = \tfrac{1}{2} \qquad \text{for every } m \geq 0.$$

A larger cutoff resolves a narrower transition, but it does not make the finite trigonometric polynomial discontinuous. The two displayed cutoffs retain 11 and 31 signed frequency bins, respectively, not five and fifteen coefficients.

An exact known-packet comparison

Take $s(t) = \frac{1}{4}\cos(2\pi t) + \frac{1}{10}\cos(6\pi t)$. For $m \geq 3$ this smooth packet is already in the retained Fourier span. Thus $S_m s + j = s + j$ exactly in this special model, whereas $S_m(s + j)$ still replaces the pulse by $S_m j$. The values assigned at the two endpoints matter for pointwise error, but do not affect an L^p norm for finite p. The illustrated ringing is not a root enclosure or an asymptotic error certificate.

> **Read this as:** Increasing spectral resolution and retaining a known discontinuity are different operations, with different pointwise consequences.

MASK GEOMETRY / TRANSLATION BY ONE SIXTEENTH

Small location errors, different norms

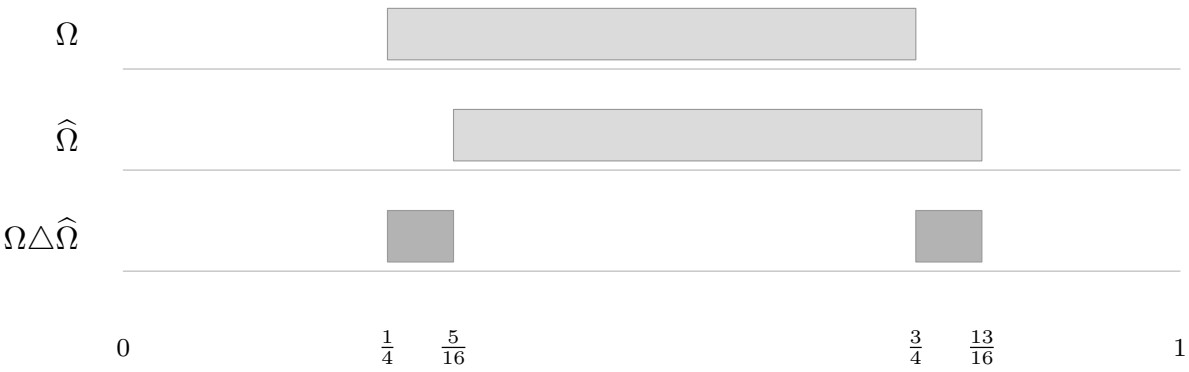
Main text: Theorem 7.1 and its norm conventions, p. 14.

In Lebesgue measure on the line, let

$$\Omega = [\tfrac{1}{4}, \tfrac{3}{4}], \qquad \widehat{\Omega} = [\tfrac{5}{16}, \tfrac{13}{16}], \qquad a = \widehat{a} = 3.$$

The two intervals have the same length. Their disagreement consists of one missed strip and one extra strip, each of width 1/16.

$$\delta = 1/16, \qquad \mu(\Omega \triangle \widehat{\Omega}) = 2\delta = 1/8$$



$$\|j - \widehat{j}\|_1 = 3/8, \qquad \|j - \widehat{j}\|_2^2 = 9/8, \qquad \|j - \widehat{j}\|_\infty = 3$$

$$|\Omega \triangle \widehat{\Omega}| = \tfrac{1}{8}, \qquad j - \widehat{j} = 3(\mathbf{1}_\Omega - \mathbf{1}_{\widehat{\Omega}}).$$

error measure	L^1 norm	squared L^2 norm	L^∞ norm
exact value	3/8	9/8	3

Area shrinks; the height of the missed jump does not

For a translation by δ with $0 < |\delta| < 1/2$, the symmetric difference has measure $2|\delta|$. At finite p,

$$\|j - \widehat{j}\|_p = 3(2|\delta|)^{1/p}.$$

It tends to zero as the displacement tends to zero. The essential supremum stays 3 for every nonzero displacement, because the error reaches height 3 on a set of positive measure. At exactly zero displacement all norms vanish. A nonempty disagreement set of measure zero would be different: it can affect the pointwise supremum while leaving the essential norm unchanged. This is why the paper distinguishes the two statements.

> **Read this as:** A shrinking boundary strip can give small integrated error and still contain a full-height jump error.

COUNTING MEASURE / AN EXACT FINITE BUDGET

Separate amplitude error from mask error

Main text: Theorem 7.1 and the background-error inequality, p. 14.

On the 16 sample addresses, let $\Omega = \{6, \dots, 11\}$ and $\widehat{\Omega} = \{7, \dots, 12\}$. Use true amplitude $a = 3$ and estimate $\widehat{a} = 5/2$. There are six true samples, five overlapping samples, and two disagreement samples.

$$a = 3, \quad \widehat{a} = 5/2, \quad |\Omega| = 6, \quad |\Omega \triangle \widehat{\Omega}| = 2$$

	4	5	6	7	8	9	10	11	12	13	14
j	0	0	3	3	3	3	3	3	0	0	0
$\widehat{j}$	0	0	0	$\frac{5}{2}$	$\frac{5}{2}$	$\frac{5}{2}$	$\frac{5}{2}$	$\frac{5}{2}$	$\frac{5}{2}$	0	0
$j - \widehat{j}$	0	0	3	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{5}{2}$	0	0

$$\|j - \widehat{j}\|_1 = 3 + 5(1/2) + 5/2 = 8$$

The residual splits exactly as

$$3\mathbf{1}_\Omega - \tfrac{5}{2}\mathbf{1}_{\widehat{\Omega}} = \tfrac{1}{2}\mathbf{1}_\Omega + \tfrac{5}{2}(\mathbf{1}_\Omega - \mathbf{1}_{\widehat{\Omega}}).$$

The missed sample has error 3, the five shared samples have error $1/2$, and the extra sample has error $-5/2$. With *counting measure*,

$$\|j - \widehat{j}\|_1 = 8, \qquad \|j - \widehat{j}\|_2^2 = \tfrac{33}{2}, \qquad \|j - \widehat{j}\|_\infty = 3.$$

Read the theorem as a budget, not an equality claim

Here the L^1 bound is attained:

$$|3 - \tfrac{5}{2}|\,|\Omega| + \tfrac{5}{2}|\Omega \triangle \widehat{\Omega}| = \tfrac{1}{2} \cdot 6 + \tfrac{5}{2} \cdot 2 = 8.$$

The L^2 triangle bound is $\frac{1}{2}\sqrt{6} + \frac{5}{2}\sqrt{2}$, and need not be sharp. A background estimate adds another term $\|s - \widehat{s}\|_p$; cancellation between terms can make the total smaller than the bound. Normalized RMSE divides the L^2 norm by $\sqrt{16}$. A boundary-band RMSE uses the band size instead. Changing that denominator changes the reported quantity.

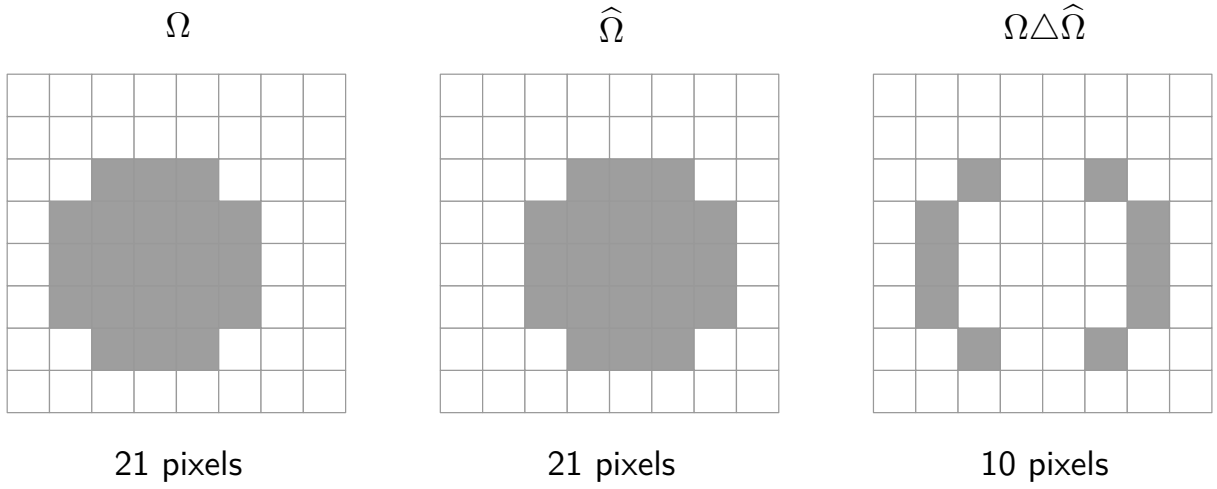
> **Read this as:** Mask error, amplitude error, and background error are separate mathematical inputs. A single reconstruction score conceals that separation.

A FINITE IMAGE / AN AMBIGUOUS OBSERVATION

Geometry localizes the error; data may not identify the geometry

Main text: Sections 7–9 and Algorithm A.3, pp. 14–19 and 34.

On the integer 8×8 grid, choose the disk-shaped mask $\Omega = \{(i, j) : (i-3)^2 + (j-3)^2 \leq 5\}$ and translate it one pixel to the right. Each mask contains 21 pixels; their symmetric difference contains exactly 10.



A geometric error calculation is not a segmentation algorithm

For a common amplitude a, the squared counting-measure error is $10a^2$, entirely on the disagreement pixels. The full-image mean squared error is $10a^2/64$. Both are exact, but answer different normalization questions. Now consider the four-sample observation

$$y = (0, \tfrac{3}{2}, 3, 3).$$

It is consistent with either true signal

$$u = (0, 0, 3, 3), \qquad v = (0, 3, 3, 3)$$

under deterministic sup-norm noise at most $3/2$. Indeed $y = (u + v)/2$, and both error norms equal $3/2$. A rule receiving only y and that noise bound cannot be guaranteed to select both distinct true masks correctly.

What additional structure would have to do

A lower noise level, separation, contrast, geometry, or a specified prior can change the identification problem. The example rules out a guarantee at the *stated* noise radius and signal pair, not all recovery results. The manuscript's positive-contrast estimator is deliberately narrower than a universal segmentation method.

> **Read this as:** Exact synthesis can preserve an identified boundary. It cannot decide between two boundaries that the observation model makes indistinguishable.

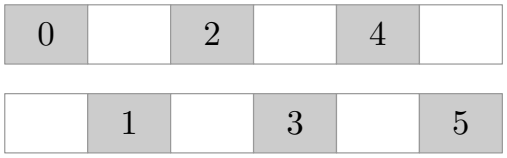
POINTWISE SIGNALS VERSUS CAUCHY COEFFICIENTS

Two products, two packet laws

Main text: Theorem 12.1 and its formal-series conventions, p. 25.

The word "product" must be typed. Multiplying two signal values at the same time address is not the same operation as multiplying generating series. The distinction changes which packets can interact.

POINTWISE SIGNAL PRODUCT



$$(P_{2,0}x)\,(P_{2,1}y) = 0$$

CAUCHY COEFFICIENT PRODUCT

0	1	2
1	2	0
2	0	1

output label $(i + j) \bmod 3$

$$A = 1 + 2t + 3t^2,\ B = 2 + t^2 : \qquad P_{3,0}(AB) = 2 + 2t^3$$

Pointwise multiplication preserves the same address

For time-residue masks, $(P_{N,i}x)[n](P_{N,j}y)[n] = 0$ when $i \neq j$. Therefore

$$P_{N,k}(x \cdot y) = (P_{N,k}x) \cdot (P_{N,k}y).$$

Here the dot means multiplication at the same sample, not convolution.

Cauchy multiplication adds degrees

For formal series $A(t), B(t)$, the coefficient of t^n is a sum over $a + b = n$. Consequently the output residue is $i + j \pmod N$:

$$P_{N,k}(AB) = \sum_{i+j\equiv k\ (N)} (P_{N,i}A)(P_{N,j}B).$$

For $N = 3$, take $A = 1 + 2t + 3t^2$ and $B = 2 + t^2$. Direct multiplication gives

$$AB = 2 + 4t + 7t^2 + 2t^3 + 3t^4, \qquad P_{3,0}(AB) = 2 + 2t^3.$$

The $2t^3$ term comes from residues $1 + 2$, not $0 + 0$. At power zero, the empty product contributes 1 only in residue zero. These are formal coefficient identities and require no convergence assumption.

> **Read this as:** Before applying a packet multiplication rule, specify whether multiplication acts on sample values or on series coefficients.

FULL INPUT TUPLES / EXACT CYCLIC INDEXING

Nonlinear neighbors cross channel boundaries

Main text: The nonlinear shifted-filter example, p. 25.

For the periodic running signal define

$$F(x)[n] = x[n] + \tfrac{1}{2}x[n]x[n-1].$$

This is nonlinear and not a convolution operator. Nevertheless its two-channel representation is exact, provided the neighbor comes from the correct channel and index.

$$F(x)[n] = x[n] + \tfrac{1}{2}x[n]x[n-1]$$

$$F_0[n] = e[n] + \tfrac{1}{2}e[n]o[n-1] \qquad F_1[n] = o[n] + \tfrac{1}{2}o[n]e[n]$$

$$e[1] = 1/2, \quad o[0] = 1/4 \qquad o[1] = 1/4, \quad e[1] = 1/2$$

$$F(x)[2] = 9/16 \qquad F(x)[3] = 5/16$$

phase wrap changes the neighbor address; it does not remove the mixed term

Put $e[q] = x[2q]$ and $o[q] = x[2q+1]$, with channel indices taken modulo eight. Then

$$F(x)[2q] = e[q] + \tfrac{1}{2}e[q]o[q-1],$$
$$F(x)[2q+1] = o[q] + \tfrac{1}{2}o[q]e[q].$$

The even output needs the *previous* odd sample; the odd output needs the current even sample. This is the same carry in the address system that appeared in the one-sample delay matrix.

Check two actual outputs

The running values are $x[1] = 1/4$, $x[2] = 1/2$, and $x[3] = 1/4$, so

$$F(x)[2] = \tfrac{1}{2} + \tfrac{1}{2}\tfrac{1}{2}\tfrac{1}{4} = \tfrac{9}{16}, \qquad F(x)[3] = \tfrac{1}{4} + \tfrac{1}{2}\tfrac{1}{4}\tfrac{1}{2} = \tfrac{5}{16}.$$

Applying the same formula independently within each decimated array would use the wrong neighbors. Keeping the full channel tuple avoids that loss of information. The new checks compare every output sample in the two implementations using exact rational arithmetic. No convergence, contraction, or approximation property follows merely from writing these exact coordinates. Those require hypotheses on the nonlinear map and the problem being solved.

> **Read this as:** Exact packet coordinates remain available for nonlinear maps. They do not turn cross-channel interactions into independent channel problems.

THE CAYLEY CHART / EXACT QUARTER TURNS

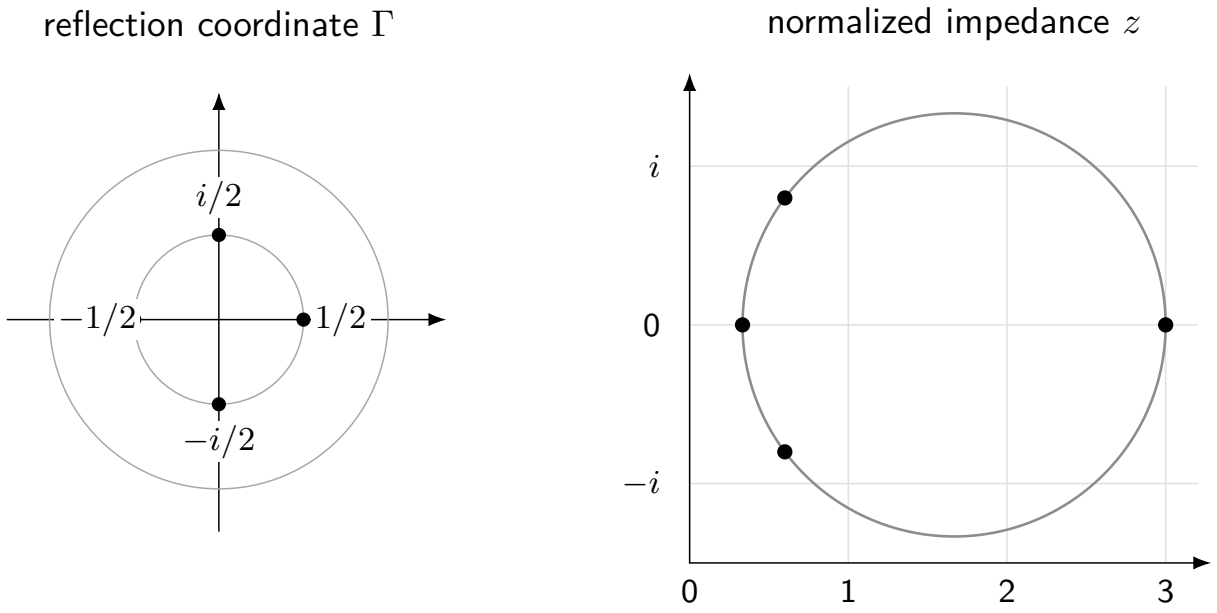
One reflection circle, four very different impedances

Main text: Theorem 13.2 and radial seeds, pp. 27–29.

Use normalized impedance z and reflection coefficient Γ:

$$\Gamma = \frac{z-1}{z+1}, \qquad z = \frac{1+\Gamma}{1-\Gamma}.$$

The circle $|\Gamma| = 1/2$ is a constant-mismatch orbit. Its four quarter-turn points map to the exact impedances shown below.



$$3, \quad \tfrac{3}{5} + \tfrac{4}{5}i, \quad \tfrac{1}{3}, \quad \tfrac{3}{5} - \tfrac{4}{5}i$$

Γ	$1/2$	$i/2$	$-1/2$	$-i/2$
z	3	$3/5 + 4i/5$	$1/3$	$3/5 - 4i/5$

The coordinates preserve structure, not Euclidean distances

A quarter turn is a rotation in the reflection disk. After pulling back, it is a Möbius map, not a Euclidean rotation of the impedance plane. Equal reflection magnitudes need not give equal impedance magnitudes. The opposite points $1/2$ and $-1/2$ become 3 and $1/3$. The continuous impedance locus is the circle $(\operatorname{Re} z - 5/3)^2 + (\operatorname{Im} z)^2 = 16/9$, also drawn in the plate. Finite passive impedances with $\operatorname{Re} z \geq 0$ map to the closed disk except $\Gamma = 1$. That missing point is obtained only by adjoining impedance infinity. The interior maps to the open right half-plane. The classical Smith grid supplies the geometry; the cyclic labels specify exact rotations on it. Neither a new physical law nor a new optimal sampling theorem is implied by this coordinate description.

> **Read this as:** A simple action in one chart can be nonlinear in another. Conjugacy preserves the action law, not the apparent shape of a displacement.

HOMOGENEOUS COORDINATES / NO DIVISION BY ZERO

A pole is a chart boundary, not a broken group action

Main text: Theorem 13.2 and Corollary 13.3, p. 27.

Represent impedance by a nonzero pair $[u : v]$, where $[cu : cv] = [u : v]$ for $c \neq 0$. A finite impedance is $[z : 1]$; infinity is $[1 : 0]$. For a nonzero reflection multiplier w, use

$$A(w) = \frac{1}{2}\begin{pmatrix} 1+w & 1-w \\ 1-w & 1+w \end{pmatrix}.$$

$$A(i) = \tfrac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$$



$z = 0$	$z = -i$	$z = \infty$	$z = i$
$\Gamma = -1$	$\Gamma = -i$	$\Gamma = 1$	$\Gamma = i$

$A(i)^4 = I$: the fourth step returns to $[0 : 1]$

The action is defined before a ratio is taken

The determinant is w, so the matrix cannot send a nonzero pair to zero. Direct multiplication gives

$$A(w)A(w') = A(ww'), \qquad A(1) = I.$$

For $w = i$, the orbit through zero is

$$[0 : 1] \longmapsto [-i : 1] \longmapsto [1 : 0] \longmapsto [i : 1] \longmapsto [0 : 1].$$

The third point is a legitimate projective point, even though a finite impedance formula has a zero denominator there. The second iterate is $A(-1)$, which swaps the two coordinates. Thus a half turn exchanges 0 and ∞ and sends each finite nonzero z to $1/z$. The matched point $\Gamma = 0$, equivalently $z = 1$, is fixed by every phase multiplier. It is not part of the boundary orbit above. The exact checks test homogeneous pairs and their rescaling equivalence, rather than introducing an artificial complex value for $1/0$.

> **Read this as:** Compute with homogeneous pairs first. A denominator becomes a chart-selection issue rather than an exception to the cyclic action.

DYADIC–ODD FACTORIZATION / TWELVE ADDRESSES

Residue labels are not impedance averages

Main text: Proposition 14.1 and Figure 15, pp. 29–30.

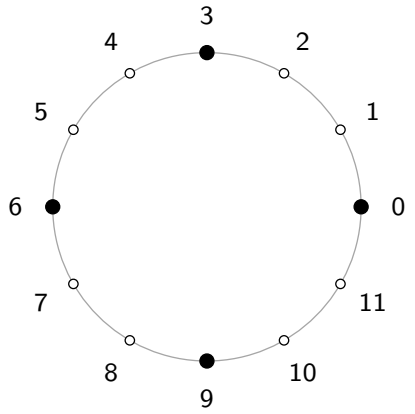
At phase order $12 = 4 \cdot 3$, the map

$$k \longmapsto (k \bmod 4,\ k \bmod 3)$$

is a bijection from twelve residue addresses to the 4×3 grid. Every cell below contains its unique original address.

	0	1	2
0	0	4	8
1	9	1	5
2	6	10	2
3	3	7	11

row $j \bmod 4$, column $j \bmod 3$



filled: the four coarse rays; open: the eight new rays

The factorization is an intersection of conditions

For integer sample addresses, including negative ones,

$$P_{12,k} = P_{4,k \bmod 4}\, P_{3,k \bmod 3}.$$

The product on the right is composition of two residue masks: both congruences must hold. Coprimality makes their intersection a single class modulo twelve. It is not a formula for multiplying or adding impedances.

Nested samples and nested projectors are different statements

The four coarse reflection rays occur at indices $0, 3, 6, 9$ of the twelve-ray grid. Thus the geometric sample set is nested. Separately, a coarse coefficient projector is the sum of its fine descendants in a common coefficient space. There is no corresponding rule saying that a coarse impedance is the average of its descendants. The inverse Cayley map is nonlinear. Even opposite reflection points have impedances 3 and 1/3 in the preceding example; their average 5/3 is not the impedance 1 of their averaged reflection coefficient zero.

> **Read this as:** CRT factors addresses. Refinement conserves coefficient packets. Neither operation authorizes linear averaging through a nonlinear chart.

LOCAL DERIVATIVES / FINITE INCREMENTS

Sensitivity grows before the pole is reached

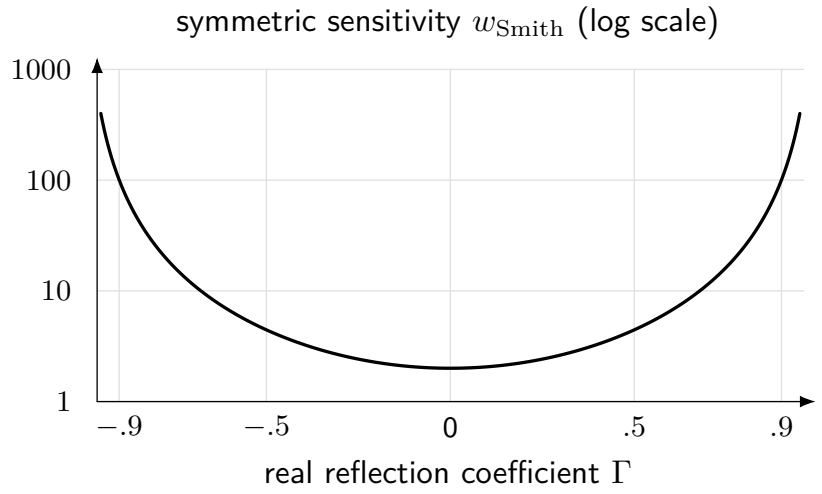
Main text: Section 14 and Boundary 14.2, p. 30.

Away from $\Gamma = 1$, the inverse Cayley map satisfies

$$\frac{dz}{d\Gamma} = \frac{2}{(1-\Gamma)^2}.$$

Along the real axis, three exact derivatives already show the scale change:

$$\Gamma = \tfrac{1}{2} : \ 8, \qquad \Gamma = \tfrac{9}{10} : \ 200, \qquad \Gamma = \tfrac{99}{100} : \ 20000.$$



Γ	$\lvert dz/d\Gamma\rvert$
$1/2$	8
$9/10$	200
$99/100$	20000

$$\Delta z = 20/9 \neq 2$$

A differential is not the full finite-error formula

For a finite increment ε, with both denominators nonzero,

$$z(\Gamma+\varepsilon) - z(\Gamma) = \frac{2\varepsilon}{(1-\Gamma-\varepsilon)(1-\Gamma)}.$$

At $\Gamma = 9/10$ and $\varepsilon = 1/100$, the actual increment is $20/9$, while derivative times increment is 2. Both are calculated exactly. Their difference is not floating-point noise; the derivative is a first-order approximation.

What a proposed weight can and cannot certify

Impedance is sensitive near $\Gamma = 1$, and admittance near $\Gamma = -1$. The symmetric weight

$$w(\Gamma) = |1-\Gamma|^{-2} + |1+\Gamma|^{-2}$$

reflects both chart sensitivities. Its growth motivates denser sampling near the endpoints, but does not establish an optimal grid, interpolation rule, or measurement allocation. A finite-error estimate must also control how close the perturbed point comes to either pole.

> **Read this as:** Use the exact finite difference for an error budget. A large derivative diagnoses conditioning; it does not by itself choose an optimal algorithm.

EVIDENCE GUIDE / THE FORMAL CONTRACT IS UNCHANGED

What the reader can now verify

Source release: Appendix D, especially the boundaries on pp. 36–39.

Three worked structures, three distinct kinds of conclusion

Structure	What is established in this appendix
Finite signal	Exact signed addressing, filter transfer, Haar reconstruction and energy, sparse coefficient error, and full-tuple nonlinear evaluation.
Finite packet tree	All 26 complete prunings, actual projector sums, the unique cost-9 optimum, and a strict perturbation-margin certificate.
Reflection orbit	Exact rational-complex coordinates, a pole-safe projective action, CRT labels, and finite sensitivity calculations.

What each evidence layer means

Manuscript theorems. The preserved v1.1.0 release maps twenty theorem-like statements to 225 supporting Lean targets in 23 proof modules. Its saved compiler and axiom records remain inherited evidence. No new Lean run is claimed for this appendix.

Exact examples. The new Python suite performs 181 checks using rational, algebraic, symbolic, and exhaustive finite computations. Seven deliberately false assertions are rejected. Numerical samples draw the continuous Fourier and sensitivity curves; they are not promoted to interval enclosures or universal error bounds.

Reproducibility. The source package retains the original manuscript, figures, Python companion, Lean sources, maps, and historical audits unchanged. The new appendix has editable LaTeX, generated TikZ plates, exact models, and isolated rebuild commands. Open `START_HERE.md`; the root runner verifies file hashes or reruns examples without invoking Lean.

The remaining boundaries are not decorative disclaimers

Known-edge synthesis requires the correct packet. A finite cost optimum depends on the supplied objective. An infinite packet tree, a universal segmentation procedure, minimax recovery rates, and optimal Smith sampling require additional theorems. The paper's two edge-recovery problems are not resolved by these examples. The 16-sample running signal is not the 1024-sample manuscript experiment; the 26-pruning example is not its 458,330-pruning benchmark. Their smaller sizes allow every relevant number to be inspected exactly.

> **Read this as:** Coordinates, approximation, identification, optimization, and physical interpretation are separate layers. The examples make their interfaces visible without erasing their hypotheses.